

Novel Graphene Hydrogel/B-Doped Graphene Quantum Dots Composites as Trifunctional Electrocatalysts for Zn–Air Batteries and Overall Water Splitting

Tran Van Tam, Sung Gu Kang, Mun Ho Kim, Seung Geol Lee, Seung Hyun Hur, Jin Suk Chung,* and Won Mook Choi*

Herein, a facile, one-step hydrothermal route to synthesize novel all-carbon-based composites composed of B-doped graphene quantum dots anchored on a graphene hydrogel (GH-BGQD) is demonstrated. The obtained GH-BGQD material has a unique 3D architecture with high porosity and large specific surface area, exhibiting abundant catalytic active sites of B-GQDs as well as enhanced electrolyte mass transport and ion diffusion. Therefore, the prepared GH-BGQD composites exhibit a superior trifunctional electrocatalytic activity toward the oxygen reduction reaction, oxygen evolution reaction, and hydrogen evolution reaction with excellent long-term stability and durability comparable to those of commercial Pt/C and Ir/C catalysts. A flexible solid-state Zn–air battery using a GH-BGQD air electrode achieves an open-circuit voltage of 1.40 V, a stable discharge voltage of 1.23 V for 100 h, a specific capacity of 687 mAh g^{−1}, and a peak power density of 112 mW cm^{−2}. Also, a water electrolysis cell using GH-BGQD electrodes delivers a current density of 10 mA cm^{−2} at cell voltage of 1.61 V, with remarkable stability during 70 h of operation. Finally, the trifunctional GH-BGQD catalyst is employed for water electrolysis cell powered by the prepared Zn–air batteries, providing a new strategy for the carbon-based multifunctional electrocatalysts for electrochemical energy devices.

energy crisis and environmental pollution has increased. Many energy systems, such as fuel cells, rechargeable metal–air batteries, and water-splitting devices, have been developed to help realize a clean energy technology. The fundamental operation of these systems primarily depends on three key electrochemical reactions of oxygen reduction reaction (ORR), oxygen evolution reaction (OER), and hydrogen evolution (HER).^[1–4] ORR and OER have been regarded as two key electrochemical processes in metal–air batteries, while OER and HER are two crucial electrochemical reactions in the water-splitting process. However, these electrochemical reactions show sluggish kinetics and high overpotentials, which limit the overall efficiency of energy devices. Therefore, noble metal and transition metal catalysts are required, including Pt for the ORR and HER, and Ir and Ru for the OER.^[5–10] However, the scarcity and high cost of precious metal-based catalysts impede the widespread technological use and large-scale application of these electrochemical energy technologies.

Therefore, extensive research has been devoted toward developing efficient and robust catalysts containing earth-abundant materials as inexpensive alternatives to noble metal catalysts, such as transition metal and carbon nanomaterial-based catalysts.^[11–19] Recently, there has been great interest in developing multifunctional (bifunctional or trifunctional) electrocatalysts with high activity toward ORR, OER, and HER due to their potentially simple fabrication and operation as well as their low production costs. For instance, multifunctional electrocatalysts for ORR and OER in metal–air batteries and for OER and HER in water-splitting devices have received considerable attention from the fields of both industrial and academic research.

In recent decades, carbon-based nanomaterials, such as carbon nanotubes (CNT) and graphene, and their composites have been studied as efficient metal-free electrocatalysts due to their stable activity and unique characteristics. In particular, carbon nanomaterials doped with N, P, S, and B atoms and their composites have demonstrated their potential as efficient multifunctional electrocatalysts by the production of numerous active sites in the carbon skeleton and the reduction of the

1. Introduction

In recent years, the demand for developing new, sustainable energy-conversion and energy-storage devices to solve the

Dr. T. V. Tam, Prof. S. G. Kang, Prof. S. H. Hur, Prof. J. S. Chung,
Prof. W. M. Choi
School of Chemical Engineering
University of Ulsan
Ulsan 44160, Republic of Korea
E-mail: jschung@ulsan.ac.kr; wmchoi98@ulsan.ac.kr

Prof. M. H. Kim
Department of Polymer Engineering
Pukyong National University
Busan 48547, Republic of Korea
Prof. S. G. Lee
Department of Organic Material Science and Engineering
Pusan National University
Busan 46241, Republic of Korea

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reaction barrier.^[20–27] Among various hetero atoms, N and B atoms have attracted much attention due to their similar atomic radii to that of carbon. For example, N-doped carbon nanomaterials have been widely exploited as metal-free electrocatalysts for ORR and OER. However, N-doped carbon materials, induced n-type transport properties, show very sensitive electronic property to the atmosphere and their electrocatalytic mechanism is still under debating. The experimental progress for B-doped carbon nanomaterials has been much less impetuous than on N-doped carbon nanomaterials because of difficulties of their synthesis and low doping concentration. The introduction of B atoms increases the hole-type charge carriers due to their lower electronegativity than carbon and the positively charged B atoms can facilitate an effective chemisorption with the negatively charged oxygen atoms. Therefore, the use of B-doped carbon nanomaterials has been regarded as an effective strategy for enhancing the electrocatalytic activity.^[18–23] The high catalytic performance of B-doped carbon nanomaterials for the ORR has been demonstrated and their bifunctional electrocatalytic activity for OER and ORR has also been shown.^[28,29] However, the development of highly active B-doped carbon nanomaterials and their catalytic mechanism still needs to be studied to develop low-cost, efficient multifunctional electrocatalysts.

Herein, we report the simple and green synthesis of novel nanocomposites consisting of a graphene hydrogel (GH) and B-doped graphene quantum dots (B-GQDs) (GH-BGQD). The materials were fabricated by a one-step hydrothermal process involving the mixture of graphene oxide (GO) and glucose with boric acid as the carbon source and a B-doping precursor for the B-GQDs. We subsequently investigated the GH-BGQD composites as metal-free carbon-based alternatives to novel metal electrocatalysts. The obtained GH-BGQD composites showed a remarkable trifunctional electrocatalytic performance toward ORR, OER, and HER with an excellent stability in alkaline media that is comparable to those of the commercial Pt/C or Ir/C electrocatalysts. A theoretical study was also performed to confirm the experimental results showing the trifunctional electrocatalytic activity of GH-BGQD. To the best of our knowledge, little work has been carried out on the trifunctional electrocatalytic activity of B-doped carbon nanomaterials. The B-GQDs containing numerous active sites were uniformly dispersed on the 3D-structured GH. These uniquely structured nanocomposites with high conductivity enable excellent access to active sites and pathways for the effective mass transport of electrolytes or gases, which result in enhanced electron transfer and diffusion distances for the ORR, OER, and HER process, facilitated by the synergic effect of the different components of the hybrid GH-BGQD nanostructure. Based on this notable trifunctional electrocatalytic behavior, we constructed Zn–air batteries using the prepared GH-BGQD material as an air electrode to achieve good performance and long-term cycle stability. Furthermore, an electrochemical water-splitting system was demonstrated using the GH-BGQD composite as both anode and cathode. This device showed a good stability, when powered by Zn–air batteries prepared using a GH-BGQD electrode. Our work provides a simple and promising approach to achieve inexpensive and highly efficient multifunctional carbon-based electrocatalysts as alternatives to novel metal catalysts.

2. Results and Discussion

2.1. Preparation and Structural Characterization of GH-BGQD Composite

There were certain difficulties related to the preparation of B-doped carbon nanomaterials, such as the need for high synthesis temperature and achieving low doping level. Therefore, a complicated fabrication process is usually required to prepare their composites with other functional materials. In this regard, we propose here a one-pot synthesis method to obtain highly porous GH-BGQD composites at low temperature. **Figure 1a** shows a schematic illustration of the key process involved in the fabrication of GH-BGQD composites. Through a one-pot hydrothermal process of the mixture of glucose, boric acid, and GO, B-GQDs were synthesized by graphitization of glucose in the presence of boric acid as a doping agent, and GO were simultaneously self-assembled to form a 3D-structured GH. Then, the synthesized B-GQDs were sequentially anchored onto the graphene sheet surface of the GH by π – π stacking, resulting in GH decorated by B-GQDs (GH-BGQD).

The morphology and microstructure of the GH-BGQD composites were characterized by scanning electron microscope (SEM) and transmission electron microscope (TEM) measurements. As shown in the SEM image of **Figure 1b**, the prepared GH-BGQD material presents a well-defined and interconnected 3D porous structure consisting of walls of graphene sheets and plenty of holes with dimension in the sub-micrometer to a few micrometers range. The obtained GH-BGQD materials exhibited a porous microstructure similar to a GH prepared without glucose and boric acid (**Figure S1**, Supporting Information). The TEM images of GH-BGQD shown in **Figure 1c,d** demonstrate that the B-GQDs were synthesized and uniformly decorated on graphene sheets of the GH without agglomeration. The size distribution of the B-GQDs was fairly uniform with an average diameter of ≈ 5 nm (**Figure S2**, Supporting Information). The lattice fringe was clearly observed in the high-resolution transmission electron microscope (HRTEM) image of single B-GQDs on a graphene sheet (**Figure 1e**) and the lattice spacing distance of B-GQDs is 0.24 nm, corresponding to the (1120) lattice of graphene, showing the high crystalline structure of the B-GQDs. The TEM image and corresponding elemental mapping images (**Figure 1f**) clearly show the uniform distribution of C, O, and B elements in the prepared GH-BGQD composite. N_2 adsorption–desorption isothermal were measured for the GH-BGQD material to characterize its porous structure (**Figure 1g**). The specific surface area and average pore size were calculated using the Barret–Joyner–Halenda (BJH) model. The specific surface areas of GH, GH-BGQD, and G-BGQD were determined to be 343.1, 320.4, and 77.4 m² g^{−1}, respectively, and the average pore of GH, GH-BGQD, and G-BGQD were 8.6, 6.9, and 10.3 nm (**Table S1**, Supporting Information). A highly porous structure with large surface area was formed in GH and GH-BGQD, while the G-BGQD sample had a very small specific surface area with poor porosity due to the stacking of graphene sheets. It can be seen that a high surface area and a porous structure of GH are also formed through the simultaneous synthesis of B-GQDs, as compared to GH material alone. This highly porous structure of GH-BGQD

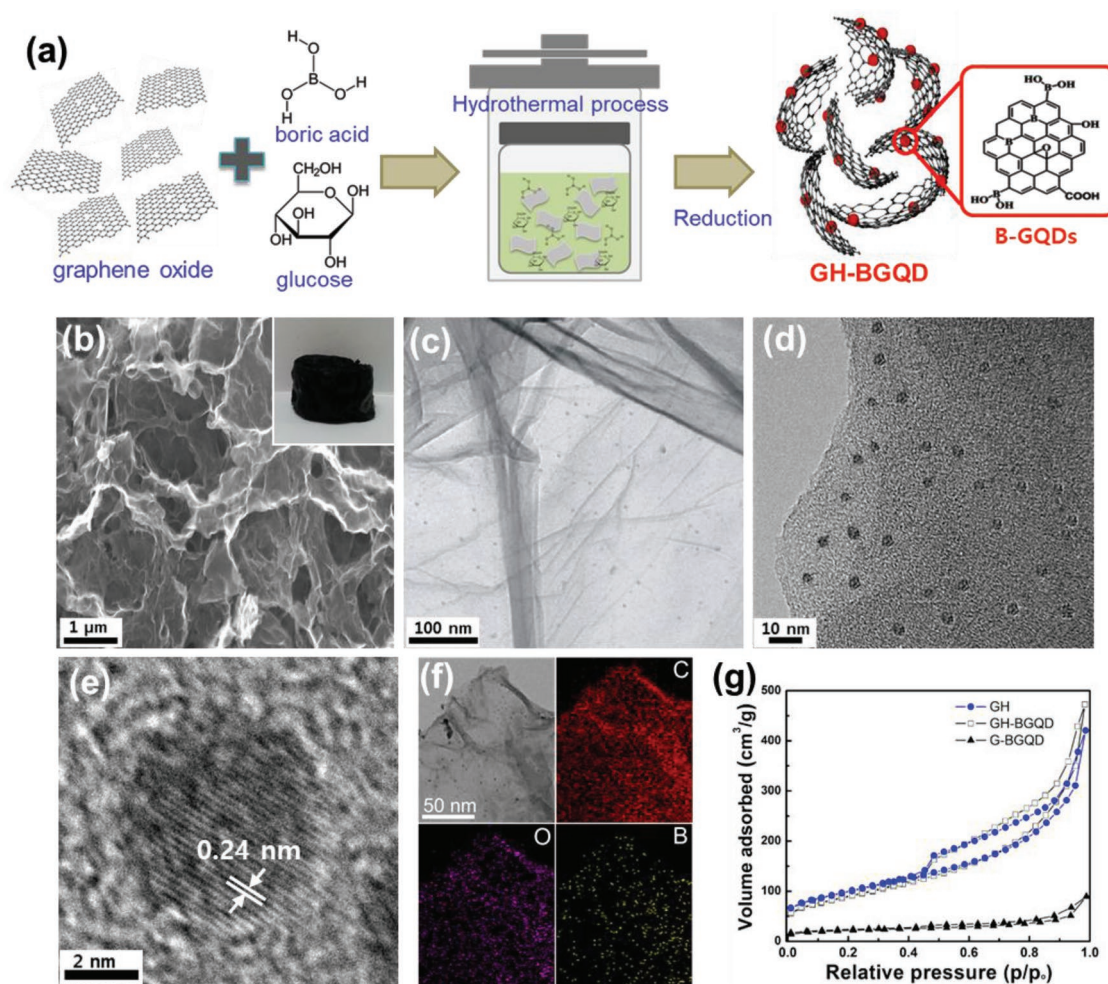


Figure 1. a) Schematic illustration of one-pot synthesis procedure for GH-BGQD. b) SEM and c,d) TEM images of GH-BGQD composite. (Inset: photograph image of the as-synthesized GH-BGQD). e) HRTEM images of B-GQDs in the GH-BGQD composite. f) TEM and corresponding elementary mapping images. g) N_2 adsorption–desorption isothermal curves of GH, GH-BGQD, and G-BGQD products.

can provide an effective diffusion path to active sites for electrolytes or intermediates, which facilitates the electrochemical reaction. In addition, Raman spectra of GH-GQD and GH-BGQD composites were compared to characterize the graphene structure and defects. Raman spectra of both samples (Figure S3, Supporting Information) exhibit two peaks at 1595 and 1350 cm^{-1} , assigned to G and D band, respectively. The G band is ascribed to sp^2 orbital of aromatic structures, while D band is corresponded to disorder or defects that existed in carbon lattice. Ratio of I_D/I_G can be used to estimate defect/disorder degree in carbon materials. The I_D/I_G ratio of GH-BGQD (0.87) is higher than that achieved from GH-GQD (0.75), reflecting more defects sites in the GH-BGQD by the introduction of B atoms in carbon skeleton, which can enhance the electrocatalytic activity.

X-ray photoelectron spectroscopy (XPS) measurements were carried out to analyze the chemical composition of the GH-BGQD composites. The XPS survey spectra of GH-BGQD and GH-GQD composite samples are given in Figure 2a. The C and O peaks assigned at 530.4 and 284.2 eV are observed in all the spectra, whereas the B peak at 190 eV is only observed

for the GH-BGQD samples. Based on the integrated peak area, the B content increases with increasing boric acid amount. The B content in GH-BGQD1, GH-BGQD2, and GH-BGQD3 was estimated to be 2.8, 4.4, and 4.8 at%, respectively (Figure S4, Supporting Information). In addition, when GH was prepared only with boric acid, its XPS survey spectrum shows no B peak (Figure S5a, Supporting Information), indicating that the B-doping occurs only during the graphitization of glucose with boric acid for B-GQDs, not in GH. As shown in Figure 2b, the high resolution C1s spectrum of GH-BGQD2 displays five different carbon-containing functional groups; four peaks centered at 284.5, 286.6, 287.3, and 288.9 are assigned to sp^2 carbon aromatic, C–O, C=O, and O–C=O bonds, respectively, and a B–C bonding peak located at 283.7 eV is observed, which is not found for GH-GQD (Figure S5b, Supporting Information), indicating the successful introduction of B atoms to the carbon network. The high-resolution B 1s spectrum for the GH-BGQD2 sample is shown in Figure 2c. The B 1s spectrum peak can be deconvoluted into four peaks, which are centered at 187.7, 189, 190.1, and 191.3 eV, corresponding to the BC_4 , graphitic-like BC_3 , BC_2O , and BCO_2 structures, respectively.^[21,29] The

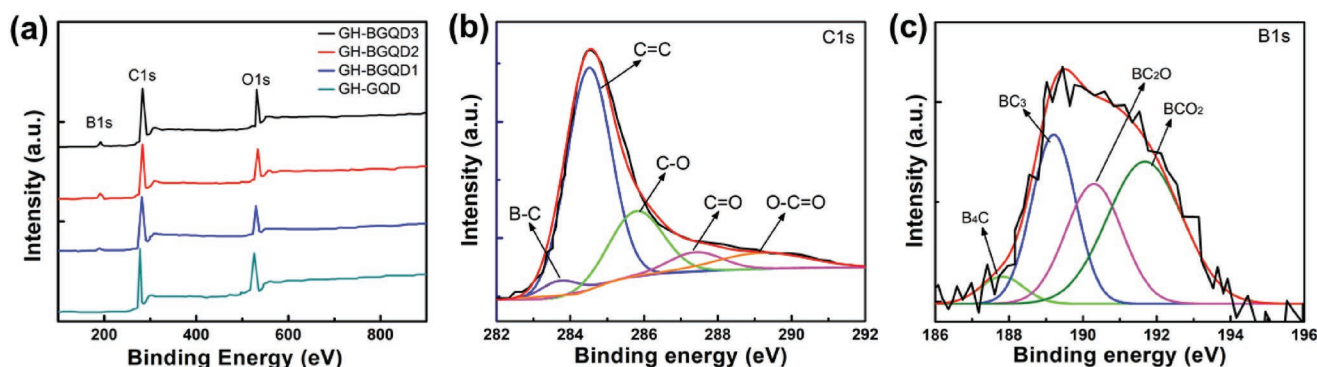


Figure 2. a) XPS survey scan of different GH-BGQD and GH-GQD composites. b) High-resolution C1s and c) B1s XPS spectra of GH-BGQD2.

calculated fractions of these four bond structures in the GH-BGQD samples are summarized in Table S2 in the Supporting Information. The GH-BGQD2 exhibits the highest content of BC_3 structure, whereas GH-BGQD3 has the highest content of the structure of boron atom bonding to carbon and oxygen, such as BC_2O and BCO_2 (Figure S6, Supporting Information). The BC_3 structure is in the basal plane species, formed by substituting a carbon atom by a boron atom without destroying the bonding configuration, where the electron of the boron atoms could be transferred to the carbon side due to their lower electronegativity. Therefore, the charge redistribution by the BC_3 structure can lead to an improved in-plane electron transfer, which could create polarized sites in the B-GQDs as active catalytic sites. The BC_2O and BCO_2 bond structures are mainly located at the edge of the B-GQDs and could decrease their electrical conductivity. However, these bonds could have better interactions with oxygen intermediates, such as H_2O and OH^- molecules which could exhibit an enhanced adsorption in the OER and HER process.^[30] In the literature, it has been reported that BCO_2 structures attached to an aromatic ring contain electrophilic boron atoms that can be suitably bonded to oxygen intermediates such as OH^- or H_2O molecules to enhance the adsorption on the catalyst in alkaline media.^[31] These various boron bond structures of the B-GQDs would be beneficial for mass transportation in an aqueous electrolyte by optimizing the ratio composition of the bonding structures for the multifunctional electrocatalyst.

2.2. Electrocatalytic Activities of GH-BGQD for ORR, OER, and HER

To evaluate the multifunctional electrocatalytic activities of the GH-BGQD composites, the ORR performance was first examined by cyclic voltammetry (CV) measurements in N_2 - and O_2 -saturated 0.1 M KOH solutions within the potential range from 0.0 to 1.2 V at a scan rate of 10 mV s⁻¹, respectively. Figure 3a depicts the CV curves for the GH-BGQD catalyst, and a featureless voltammetric current response is observed in the N_2 -saturated solution. In contrast, when the electrolyte solution is saturated with O_2 , a distinct cathodic peak is clearly observed at 0.61 V with a substantial increase of current density, which indicates that the GH-BGQD catalyst exhibits a pronounced electrocatalytic activity for the ORR. Furthermore, the linear

sweep voltammetric (LSV) measurements were performed on the prepared samples using a rotating-disk electrode (RDE) in an O_2 -saturated 0.1 M KOH solution at a rotating speed of 1600 rpm. The ORR polarization curves (Figure 3b) clearly show that the GH-BGQD material exhibits much more positive onset potentials and larger current densities than undoped GH-GQD. This improved ORR activity is attributed to the existence of B-GQDs having an abundance of active species. The polarization curves of different GH-BGQD catalysts obviously show that the ORR activity is significantly influenced by the boron doping level, with the GH-BGQD2 catalyst showing the highest onset potential (0.93 V) and the highest limiting current density (5.74 mA cm⁻²).

In addition, the G-BGQD composite exhibits lower limiting current density (2.39 mA cm⁻²) and onset potential (0.89 V) values compared to GH-BGQD2, which results from the enhanced electrolyte access to active sites by the 3D porous structure of GH-BGQD2. The ORR performance of GH-BGQD2 is comparable to that of commercial Pt/C catalysts with an onset potential of 0.93 V and the highest limiting current density (i.e., 4.03 mA cm⁻²). Furthermore, the GH-BGQD2 also possesses higher half-wave potential (0.87 V) than that of Pt/C (0.85 V), confirming the superior ORR performance of GH-BGQD2 composite. The excellent ORR performance of the prepared GH-BGQD catalyst is also compared with those of recently reported heteroatom doped carbon catalysts, as summarized in Table S3 in the Supporting Information. The RDE measurements on the GH-BGQD2 catalyst were performed at different rotating speeds and Koutecky–Levich (K–L) plots were obtained from the corresponding polarization curves (Figure S7, Supporting Information). The current densities increase simultaneously with the rotating speed due to the shortened diffusion length at higher speeds. K–L plots display a good linearity between the reserve square root of rotation speed and reserve limiting current density. In addition, the electron transfer number (n) of the GH-BGQD catalyst for the ORR was calculated from the rotating ring-disk electrode (RRDE) test (Figure S8, Supporting Information), which is between 3.73 and 3.92 in the potential range of 0.4 to 0.8 V (Figure 3c). In addition, the peroxide (H_2O_2) yield for GH-BGQD2 was calculated from RRDE measurement to study the catalytic pathway (Figure 3c). The peroxide yield for the GH-BGQD2 catalyst remained below 18% throughout the whole studied potential range, and decreased to 15% at 0.8 V. These results suggest

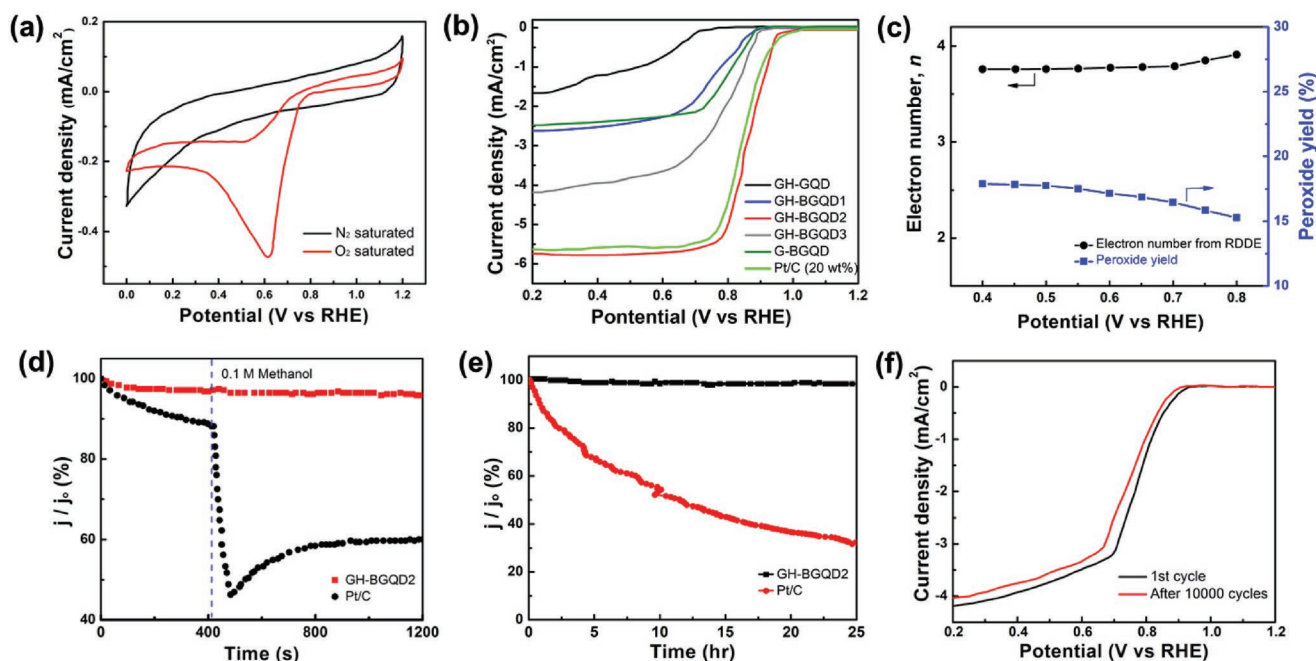


Figure 3. a) Cyclic voltammograms of GH-BGQD in N_2 - and O_2 -saturated 0.1 M KOH solution at a scan rate of 10 mV s^{-1} . b) Linear sweep voltammetry (LSV) of different GH-BGQD composites, G-BGQD, and Pt/C (20 wt%). c) The electron number (black) and corresponding peroxide yield (blue) for GH-BGQD2. d) Chronoamperometric response of GH-BGQD2 and Pt/C in 3 M methanol at a potential of 0.4 V. e) Long-time stability evaluation of GH-BGQD2 and Pt/C electrocatalysts in an O_2 -saturated 0.1 M KOH solution at 0.4 V and a rotation speed of 1600 rpm. f) Polarization ORR curves of GH-BGQD2 recorded at the first cycle and after 10 000 cycles in an O_2 -saturated 0.1 M KOH solution.

an excellent ORR catalytic activity and a four-electron reaction pathway for GH-BGQD.

Furthermore, the methanol tolerance and electrochemical stability of GH-BGQD were studied by means of a chronoamperometric test. As shown in Figure 3d, the GH-BGQD2 catalyst retained a stable current density after adding methanol to the electrolyte. In contrast, the corresponding current response on the Pt/C catalyst decreased abruptly upon methanol addition. The chronoamperometric measurement (Figure 3e) demonstrated that about 94% of the initial current density was retained for the GH-BGQD2 catalyst after 25 h of continuous operation, whereas the current density of the Pt/C catalyst substantially decayed to 38% of its initial value. Figure 3f displays the polarization curves obtained for the GH-BGQD2 catalyst after 10 000 cycles of CV scan. It can be seen that the GH-BGQD2 material only shows small decays of the onset potential and current density without any prominent deterioration. Also, the HRTEM image of GH-BGQD2 material shows that the morphology of GH-BGQD is maintained after 1000 cycles without the agglomeration and detachment of B-GQDs from GH support (Figure S9, Supporting Information). These results demonstrate the superior methanol tolerance and excellent stability of the GH-BGQD catalyst toward the ORR process in comparison to the commercial Pt/C catalyst. Furthermore, the GH-BGQD2 also shows good catalytic activity in acidic media (0.5 M H_2SO_4) with slight shift in LSV curve, compared to performance in alkaline media. After 10 000 cycles in acid media, the LSV curve of the GH-BGQD2 composite exhibits small change, indicating that the GH-BGQD2 composite also has high stability in acidic media (Figure S10, Supporting

Information). The enhanced stability of GH-BGQD catalyst is attributed to the 3D porous structure of the graphene hydrogel, which possesses superior chemical and mechanical stability and hinders the aggregation of catalyst during operation.

The GH-BGQD composites also exhibit efficient electrocatalytic activities toward the OER, which is highly useful for rechargeable metal–air batteries. We performed LSV tests in N_2 -saturated 0.1 M KOH solution using RDE measurements at a rotating speed of 1600 rpm. Figure 4a shows the OER polarization curves for GH-GQD, G-BGQD, GH-BGQD, and Ir/C (20 wt%) catalysts. It is obvious that the GH-BGQD composites exhibit lower onset potentials (e.g., 1.51 V for GH-BGQD2) comparable to undoped GH-GQD (1.79 V) but similar values compared to Ir/C catalysts (1.53 V). This enhanced OER catalytic efficiency is attributed to the effective incorporation of B-GQDs and the abundant catalytic species of GH-BGQD. Among the prepared products, the GH-BGQD2 catalyst exhibits the best OER performance, that is, an overpotential of 370 mV at current density of 10 mA cm^{-2} . This value is comparable to that reported for Ir/C catalysts (390 mV) but lower than those of reported for other carbon-based catalysts (Table S4, Supporting Information). On the other hand, the G-BGQD composite exhibits a much higher onset potential (1.68 V) and smaller current density than the GH-BGQD catalyst, suggesting that the porous structures of GH-BGQD can facilitate the effective mass transport of oxygen intermediates and oxygen gas. Figure 4b shows Tafel plots calculated from polarization curves to gain insight into the kinetic reactions. The Tafel slopes are determined to be 125, 70, and 84 mV dec^{-1} for GH-BGQD1, GH-BGQD2 and GH-BGQD3, respectively. The

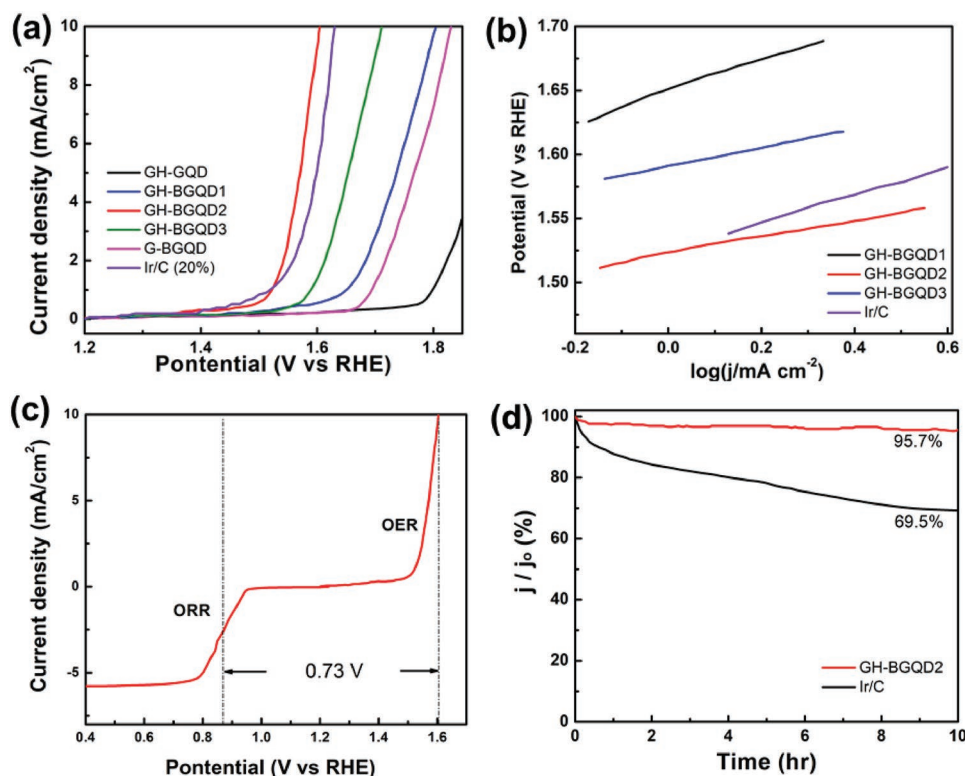


Figure 4. a) LSV OER curves for different GH-BGQD, G-BGQD, and Ir/C (20 wt%) at 1600 rpm. b) Tafel plots for GH-BGQD composites and Ir/C. c) LSV curves of GH-BGQD2 in an O_2 -saturated 0.1 M KOH solution at 1600 rpm, showing the bifunctional activities toward ORR and OER. d) Chronoamperometric response of GH-BGQD2 and Ir/C in N_2 -saturated 0.1 M KOH at 1600 rpm.

slope of GH-BGQD2 is lower than that of the commercial Ir/C catalyst, indicating that GH-BGQD2 shows the most favorable kinetic reaction toward OER. In addition, Figure 4c shows that the polarization curve in the range from 0.4 to 1.6 V clarifies bifunctional catalytic activities of GH-BGQD2 toward ORR and OER, where the potential difference ΔE between the two reactions ($\Delta E = E_{j=10} - E_{1/2}$, where $E_{j=10}$ is the operating potential to deliver 10 mA cm^{-2} for OER and $E_{1/2}$ is the half-wave potential of ORR) is determined to be 0.73 V. This value is smaller than those reported for other metal-based catalysts,^[32–34] showing the excellent bifunctional ORR and OER activities of GH-BGQD2. The chronoamperometric response of GH-BGQD2 and Ir/C catalysts was measured at a constant potential of 1.55 V for 10 h operation, as shown in Figure 4d. Remarkably, the GH-BGQD2 catalyst maintains 95.7% of its initial current density. In contrast, a larger current density loss is observed for the Ir/C catalyst. Also, the polarization curve of the GH-BGQD2 catalyst is slightly shifted to the higher potential region after 500 cycles with a negligible change in onset potential (Figure S11, Supporting Information), indicating the superior operating stability of the GH-BGQD catalyst for the OER.

Along with the excellent bifunctional catalyst performance toward ORR and OER, the GH-BGQD catalyst also exhibits an efficient HER performance. Figure 5a shows the HER polarization curves of the prepared composites and Pt/C, measured in N_2 -saturated 0.1 M KOH solutions. It can be seen that all the GH-BGQD catalysts display an enhanced HER activity due to the catalytic functional groups of B-GQDs in comparison with

the GH-GQD composite. In particular, the GH-BGQD2 catalyst delivers a higher onset potential of -0.08 V , compared to the other catalysts of GH-BGQD1 (-0.41 V), GH-BGQD3 (-0.25 V), and G-BGQD (-0.33 V). In addition, the overpotential delivered at a current density of -10 mA cm^{-2} is 130 mV for GH-BGQD2, superior to those of other GH-BGQD catalysts. Although this performance is still not as good as that of the commercial Pt/C catalyst, GH-BGQD2 shows the HER performance comparable to that of recently reported heteroatom-doped carbon-based electrocatalysts (Table S5, Supporting Information). Tafel plots were established from the polarization curves to further study the basics of the reaction kinetics, as shown in Figure 5b. Tafel slope of GH-BGQD2 is measured to be 95 mV dec^{-1} , which is smaller than that of other GH-BGQD catalysts and similar value with Pt/C catalyst. Polarization curves were also measured for the GH-BGQD2 catalyst at various scanning rates between 5 and 100 mV s^{-1} (Figure S12, Supporting Information), exhibiting stable curves without any prominent changes. Figure 5c shows the polarization curves recorded for GH-BGQD2 during first cycle and after 2000 cycles at a scan rate of 10 mV s^{-1} . The HER activity of GH-BGQD2 after 2000 cycles shows a negligible potential change around 8 mV compared to the first cycle. Additionally, chronoamperometric curves are measured for the GH-BGQD2 catalyst at a constant overpotential of 410 mV for 10 h of operation to demonstrate the long-term catalytic stability (Figure 5d). It is clearly observed that the GH-BGQD2 catalyst exhibits only a small decay of current density (6.2%) compared to commercial Pt/C catalysts (49.8%), demonstrating

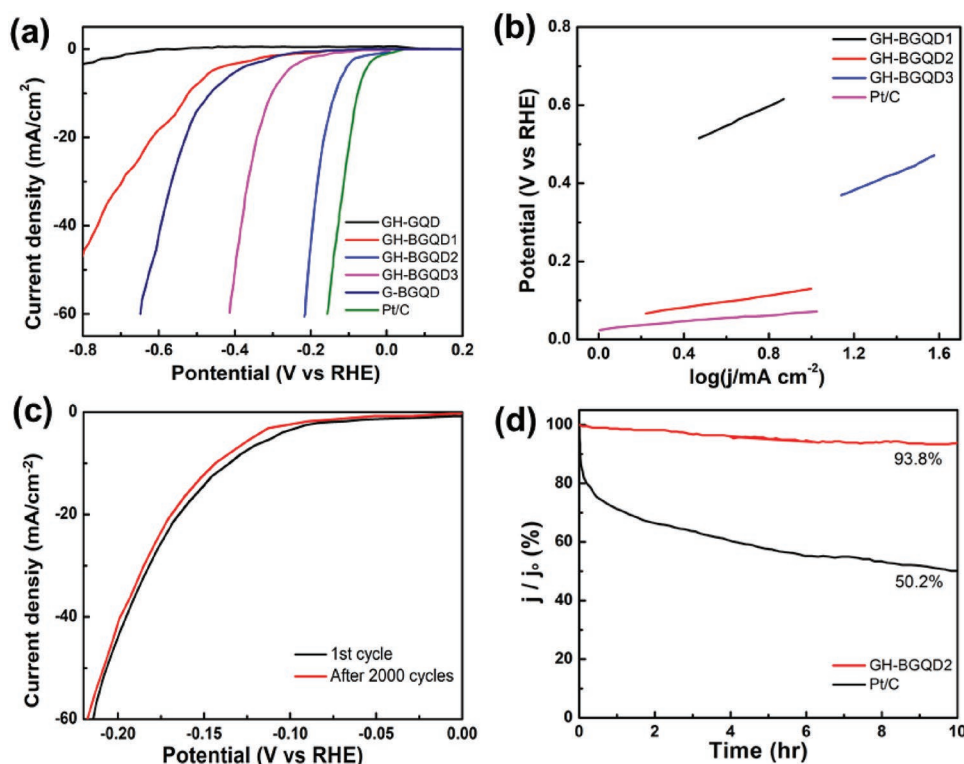


Figure 5. a) Polarization curves for different GH-BGQD composites, G-BGQD, and Pt/C (20 wt% Pt). b) Tafel plots of GH-BGQD composites and Pt/C. c) HER polarization curves for GH-BGQD2 during the first cycle and after 2000 cycles at a current density of 10 mA cm^{-2} . d) Time dependence of the current density for GH-BGQD2 at a constant overpotential of 410 mV.

the superior stability of GH-BGQD catalyst for the HER. The above-mentioned electrochemical performance results for ORR, OER, and HER suggest that the prepared GH-BGQD catalyst could be a highly efficient carbon-based trifunctional catalyst. Also, the GH-BGQD material provides important implications for developing and designing highly efficient carbon-based metal-free catalysts for various electrochemical applications with low cost, scalability, and an environmentally friendly method.

The existence of various incorporated boron functional structures may be related to the multifunctional catalytic activities of the GH-BGQD material toward ORR, OER, and HER. A previous density functional theory (DFT) study carried out in our group demonstrated that the oxygen adsorption energy on the BC_3 structure was higher than that of the other BC_2O and BCO_2 structures, resulting in a favorable adsorption on the BC_3 structure toward the ORR. In addition, we reported that the excellent electron transport feature of the BC_3 structure was also attributed to its superior ORR performance.^[35] To investigate the electrocatalytic activity toward OER and HER, the DFT method was employed to study the adsorption energy of OH^- and H_2O on different bonding structures of the B-GQDs. Figures S13 and S14 in the Supporting Information show the representative configurations and adsorption energies of OH^- and H_2O molecules on different boron-bond structures of the B-GQDs. First, the GQDs without B-doping show the negatively lower adsorption energies of both OH^- and H_2O of -0.316 and -0.159 eV, respectively, compared to B-GQDs (Figure S15, Supporting Information), indicating that B-GQDs can have the favorable

adsorption of both OH^- and H_2O on the surface. The adsorption energies of OH^- and H_2O on BCO_2 were higher than those of other BC_3 and BC_2O bonds, indicating that the adsorption of OH^- and H_2O molecules on BCO_2 is more favorable. It may be speculated that the boron atom in the BCO_2 structure is easily polarized due to the large electronegative difference, which favors the adsorption of polar OH^- and H_2O molecules. The negatively charged oxygen atom and the positively charged boron atom in BCO_2 effectively interacts with the H and O atom of OH^- and H_2O molecules, respectively, which facilitates the adsorption of OH^- and H_2O molecules to enhance the electrocatalytic activity toward OER and HER. The BCO_2 bond content of GH-BGQD2 is slightly smaller (1.24%) than that of GH-BGQD3 (1.39%), which indicates that the GH-BGQD2 composite can also have favorable adsorption of OH^- and H_2O molecules as the GH-BGQD3.

Furthermore, electrochemical impedance studied was carried out on the GH-BGQD composites. The Nyquist curves obtained for all the products exhibited a semicircle in the low frequency region with the GH-BGQD2 composite exhibiting the smallest semicircle and the lowest charge transfer resistance value (R_{ct}) among them (Figure S16, Supporting Information). This result is attributed to much higher contents of graphite-like BC_3 functional groups in the GH-BGQD2 composite (2.07%), compared to that of GH-BGQD3 (0.98%). The graphite-like BC_3 structure can increase the electron density of the carbon atoms due to the higher electronegativity of the boron atom, thus enhancing the electrical conductivity of the B-GQDs, whereas the oxygenated

boron structures BCO_2 and BC_2O often lead to a low electrical conductivity. These results are also confirmed by measurements of the electrical conductivity of the composite products (Figure S17, Supporting Information). The GH-BGQD2 exhibits a much higher electrical conductivity than the other composite products. Therefore, the superior R_{ct} and electrical conductivity of GH-BGQD2 facilitate a fast charge transfer between the electrode and the electrolytes for an efficient catalytic performance, which is consistent with the superior electrocatalytic performance of GH-BGQD2, as described above. These results can elucidate that the excellent trifunctional electrocatalytic activity of GH-BGQD2 is ascribed to the synergetic combination of the favorable adsorption capability of the BCO_2 bond and the superior electrical conductivity from the BC_3 bond.

2.3. Zn–Air Battery and Overall Water Splitting Using GH-BGQD

To demonstrate the excellent catalytic activity of the GH-BGQD2 composite for both ORR and OER, a solid-state Zn–air battery with a two-electrode configuration was constructed. The prepared battery consisted of GH-BGQD2 loaded on a Ni foam as the air electrode, a Zn foil as the anode, and a polyvinyl alcohol (PVA) hydrogel film containing 2.0 M KOH and 0.2 M ZnCl_2 as the solid electrolyte and membrane (Figure 6a). For comparison, a commercial Pt/C+Ir/C catalyst (weight ratio is 1:1) was used as the air electrode for a control electrode. The open-circuit potential was ≈ 1.40 V for the GH-BGQD2 electrode (Figure S18a, Supporting Information), higher than that for the Pt/C+Ir/C electrode (1.36 V), confirming its excellent

catalytic performance. The GH-BGQD2 electrode delivered discharge voltage of 1.23 V at a current density of 5 mA cm^{-2} and maintained it for 25 h of discharge without significant voltage drop (Figure S18b, Supporting Information). Figure 6b shows the changes in discharge voltage with time at different current densities of 2, 5, and 10 mA cm^{-2} . The discharge voltage of the GH-BGQD2 electrode decreased from 1.28 to 1.23 and 1.18 V for current densities ranging from 2 to 5 and 10 mA cm^{-2} . Also, the discharge voltage recovered to 1.23 V when the current density decreased to 5 mA cm^{-2} , indicating a good stability of the GH-BGQD2 electrode. Figure 6c depicts the discharge polarization and power density curves for the Zn–air batteries. The GH-BGQD2 electrode delivered a slightly higher discharge voltage (0.98 V at 50 mA cm^{-2}), compared to the Pt/C+Ir/C electrode (1.00 V at 50 mA cm^{-2}). The peak power density of the GH-BGQD electrode was 112 mW cm^{-2} with a current density of 150 mA cm^{-2} , which outperforms the Pt/C+Ir/C electrode (95 mW cm^{-2} and 132 mA cm^{-2}). The specific capacity, normalized to the mass of consumed Zn, was calculated to be 687 mAh g^{-1} at a current density of 10 mA cm^{-2} , corresponding to a high energy density of 810 Wh kg^{-1} (Figure 6d). This promising performance of the GH-BGQD2 electrode was attributed to its porous structure and the synergetic effects of the B-GQDs and the GH substrate, which facilitates the diffusion of O_2 gas and electrolyte to the catalytic sites. Galvanostatic charge and discharge cycling tests were performed at a current density of 5 mA cm^{-2} with a cutoff time of 20 min per cycle (Figure 6e). Throughout a continuous 300 cycles (100 h), the GH-BGQD electrode showed a stable discharge voltage of 1.23 V with a negligible round-trip overpotential change. Similarly, no significant

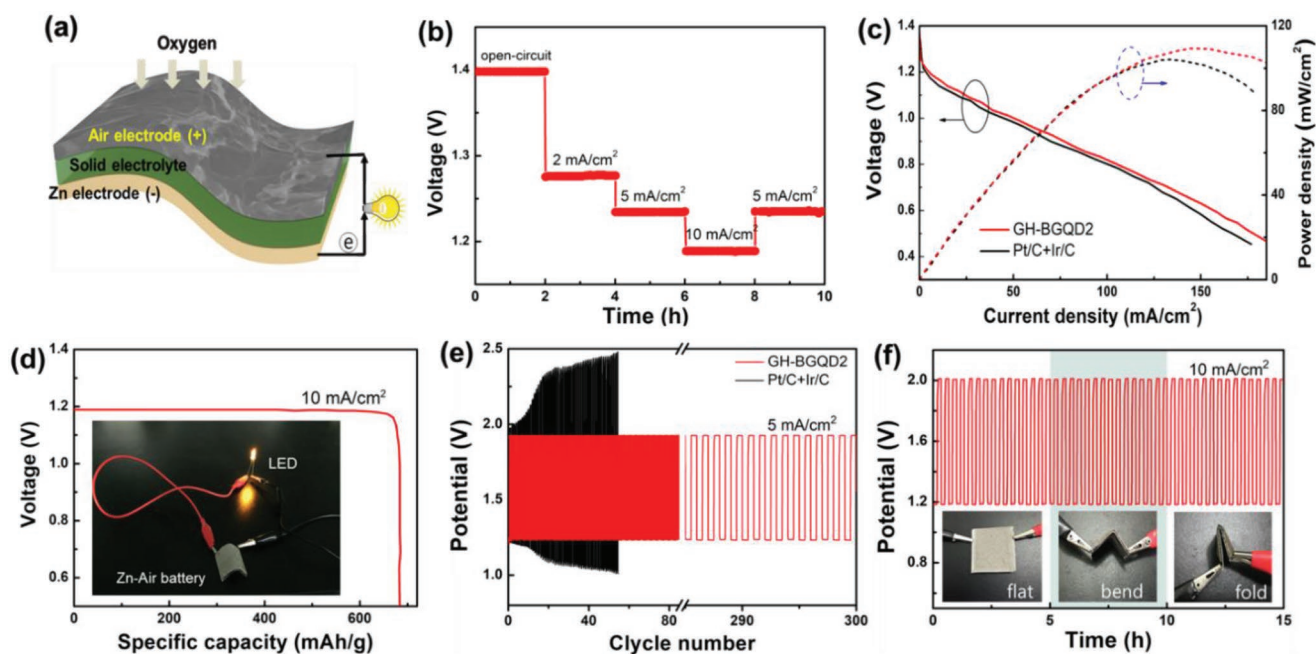


Figure 6. a) Schematic configuration of the solid-state Zn–air battery studied herein. b) Discharge curves of Zn–air battery fabricated using the GH-BGQD2 composite as the air electrode at various current densities. c) Polarization curves and corresponding power density plots. d) Specific capacity of the GH-BGQD based Zn–air battery at a current density of 10 mA cm^{-2} . e) Discharge/charge cycling curves for the GH-BGQD based Zn–air battery at a current density of 5 mA cm^{-2} (20 min per cycle). f) Discharge/charge cycling curves for the GH-BGQD based Zn–air battery at a current density of 10 mA cm^{-2} under different bending states. The inset of (d) shows a photograph of a LED ($\approx 1.0 \text{ V}$) powered by the GH-BGQD based Zn–air battery with bending state.

overpotential changes were also observed in cycling tests performed at a higher current density of 10 mA cm^{-2} (Figure S19, Supporting Information), indicating the excellent long-term cycling stability of the GH-BGQD2 based Zn-air battery. In contrast, the Pt/C+Ir/C electrode exhibited a poor cycling stability with a sharp increase in overpotential along 55 cycles (19 h), due to the agglomeration and detachment of metal nanoparticles from carbon support. Furthermore, we evaluated the mechanical stability of the solid-state Zn-air battery using the GH-BGQD2 electrode. The inset image of Figure 6d shows that an orange light-emitting diode (LED, 1.0 V) is continuously lit by the GH-BGQD2 based Zn-air battery under bending condition. In addition, the solid-state battery with GH-BGQD2 electrode displayed stable charge (2.01 V) and discharge potentials (1.18 V) at a current density of 10 mA cm^{-2} for 15 h without apparent potential degradation, even when bent to different angles, suggesting a good flexibility and cycle stability of the GH-BGQD2 based Zn-air battery (Figure 6f).

Considering its excellent OER and HER catalytic activity, we further explored the GH-BGQD2 electrode in a two-electrode overall water splitting system, where the GH-BGQD2 electrodes loaded on Ni foams were used for both the anode and cathode in a 0.1 M KOH solution. The bifunctional activity for OER and HER was investigated and in each case, the polarization curves of the GH-BGQD2 electrode exhibited a voltage difference of 1.52 V at a current density of 10 mA cm^{-2} between the HER and OER processes (Figure S20, Supporting Information). The water electrolysis cell built using the GH-BGQD2 composite achieves the current density of 10 mA cm^{-2} at a cell voltage of only 1.61 V with apparent H_2 and O_2 bubbles

generated on the surface of both electrodes, which outperforms that of Pt/C and Ir/C electrodes (1.8 V) (Figure 7a). In addition, the chronopotentiometric (at current density of 10 mA cm^{-2}) and chronoamperometric (at a voltage of 1.61 V) curves of the GH-BGQD2 cell exhibited negligible changes over 70 h of operation. These results demonstrate that the GH-BGQD2 electrode is an efficient and durable electrocatalyst for practical water electrolysis applications. For further practical applications of the trifunctional catalytic performance, we integrated the water electrolysis cell with Zn-air batteries using GH-BGQD2 electrodes, as presented in Figure 7c and Video S1 in the Supporting Information. The two-electrode water electrolysis cell was powered by two Zn-air batteries connected in series using the GH-BGQD2 material as both cathode/anode electrodes for the water electrolysis cell and as the air electrode for the Zn-air batteries. During the water electrolysis driven by the two Zn-air batteries, the vigorous generation of O_2 and H_2 bubbles was clearly observed on both GH-BGQD2 electrodes, showing the successful operation of water electrolysis powered Zn-air batteries. These results confirm that the trifunctional electrocatalytic performance of the GH-BGQD electrode makes it a promising material for practical electrochemical devices.

3. Conclusion

In summary, we have demonstrated a facile and green method for the synthesis of the GH-BGQD composites as highly efficient trifunctional electrocatalysts. The prepared GH-BGQD composites exhibit outstanding trifunctional electrocatalytic

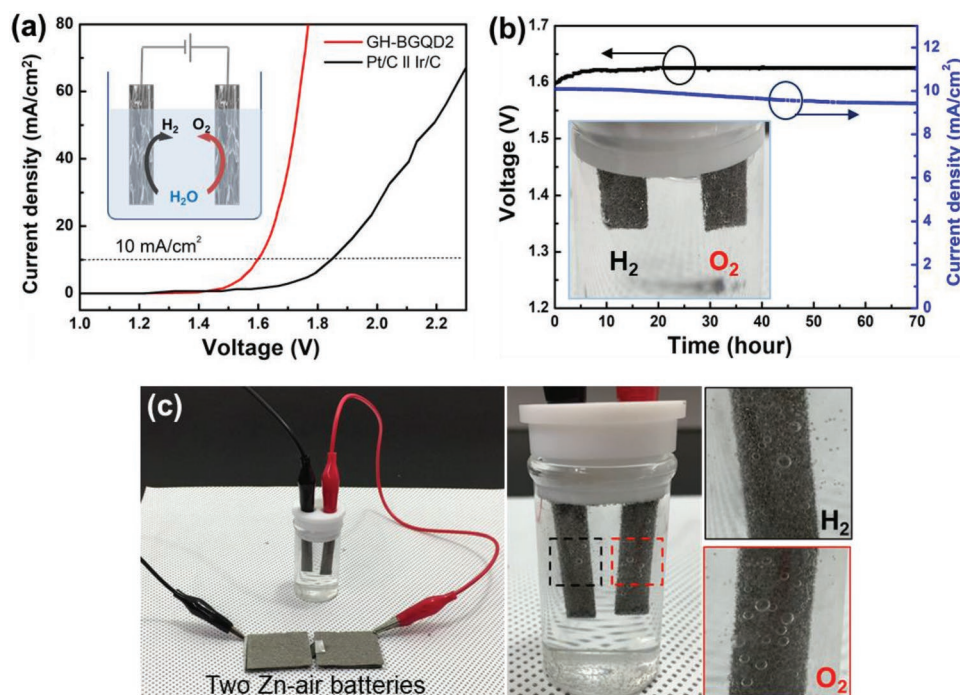


Figure 7. a) Polarization curve for water electrolysis using the GH-BGQD2 material as both anode and cathode in a 0.1 M KOH solution. b) Chronopotentiometric (at 10 mA cm^{-2}) and chronoamperometric (at 1.61 V) curves of the GH-BGQD based water electrolysis for 70 h. The inset of (b) shows a photograph of H_2 and O_2 bubbles generated on both GH-BGQD electrodes. c) Photographs of a water electrolysis cell powered by two Zn-air batteries in series and the generation of H_2 and O_2 bubbles on both GH-BGQD electrodes.

activities toward ORR, OER, and HER, which are comparable to those of the commercial Pt/C and Ir/C catalysts. Additionally, the favorable kinetic reaction with excellent stability and durability provides a promising all carbon-based electrocatalyst for replacing novel metal-based catalysts. The superior trifunctional performance of the GH-BGQD material is attributed to the efficient catalytic sites of the B-GQDs and the 3D architecture of GH, which provides easy access to active sites as well as high-rate charge and mass transport. Moreover, a Zn–air battery and a water electrolysis cell were demonstrated using the GH-BGQD electrodes and were found to deliver a superior performance with a highly stable operation. Our GH-BGQD composites contribute to a new strategy of designing efficient metal-free, carbon-based multifunctional electrocatalysts for replacing the conventional noble metal catalysts.

4. Experimental Section

Synthesis of GH-BGQD Composites: To synthesize the GH-BGQD composite, glucose (120 mg), boric acid (10 mg), and H₂SO₄ (100 μ L) were dissolved in deionized (DI) water (10 mL) upon sonication for 20 min. Then, an aqueous GO solution (50 mL, 3 mg mL^{−1}) was slowly added to the above solution with vigorous stirring for 30 min. After transferring this mixture to a 100 mL Teflon-lined stainless-steel autoclave, the hydrothermal reaction was run at 180 °C for 24 h. The resulting product was collected and soaked in DI water for 12 h to remove impurities. For the reduction of GO, the obtained composites were immersed in an HI solution (55%) in a closed vial and kept at 40 °C for 5 h. The final product was then collected and washed with plenty of DI water and finally dried in a vacuum oven at 90 °C for 1 h. The as-prepared composite samples with different boric acid concentrations of 0, 1.8, 3.6, and 5.4 wt% were referred to as GH-GQD, GH-BGQD1, GH-BGQD2, and GH-BGQD3, respectively. As a control sample, a composite of reduced graphene oxide and B-GQDs (G-BGQD) was also prepared by simply mixing GO with B-GQDs (3.6 wt% boric acid), followed by reduction using an HI solution.

Characterizations: The surface morphologies of the GH-BGQD composites were examined using a field-emission scanning electron microscope (FE-SEM, JEOL, JSM-6500F) and their nanostructures were observed using a transmission electron microscope (JEOL, JEM-2100F). UV–vis absorption spectra were collected using a Specord 210 UV–vis spectrophotometer. X-ray photoelectron spectroscopy (Thermo Fisher) was performed using monochromatic AlK α radiation (h ν = 1486.6 eV). The specific surface areas of the prepared samples were calculated by Brunauer–Emmett–Teller (BET) analysis of the nitrogen adsorption measurements (Micromeritics, ASAP 2020, USA). The electrical conductivities of the obtained products were determined using a conventional four probe Keithley 4200 semiconductor characterization system.

Electrochemical Characterizations: Electrochemical measurements were performed in a three-electrode system using a multichannel potentiostat-galvanostat analyzer (VSP, Bio-logic) at room temperature. Typically, a graphite rod was used as the counter electrode for the HER and ORR, whereas a Pt wire was used as the counter electrode for the OER. A saturated calomel electrode (SCE) was used as the reference electrode for all reactions. The potentials obtained here were converted to the reversible hydrogen electrode (RHE) scale. For the working electrodes, the catalyst active materials were prepared by adding GH-BGQD nanocomposites (5 mg) to a mixture of isopropyl alcohol (4.975 mL) and Nafion binder solution (25 μ L, 0.5 wt%, DuPont), followed by sonication for 30 min to obtain a homogeneous solution. Then, the prepared catalyst paste was loaded on rotating-disk electrode with glassy-carbon electrode (GCE, 4 mm in diameter) and Pt ring (5.5 mm inner diameter and 7.2 mm outer diameter). For

comparison, Pt/C (20% Pt on Vulcan XC72) and Ir/C (20% Ir on Vulcan XC-72) electrodes were also prepared in the same configuration. The electrochemical measurements were carried out in O₂ or N₂-saturated 0.1 M KOH electrolyte solutions at room temperature. A rotating ring-disk experiment was performed at 1600 rpm in an O₂-saturated 0.1 M KOH electrolyte solution. The number of electrons transferred (n) per oxygen molecule in the ORR process was determined by the Koutecky–Levich equation given below

$$\frac{1}{j} = \frac{1}{j_k} + \frac{1}{B\omega^{0.5}} \quad (1)$$

where j_k is the kinetic current density and ω is the angular velocity of the electrode. B could be determined from the slope of the K–L plots based on the following Levich equation

$$B = 0.2 nF(D_{O_2})^{2/3} \nu^{-1/6} C_{O_2} \quad (2)$$

where n is the number of electrons transferred per oxygen molecule, F is the Faraday constant (96 485 C mol^{−1}), D_{O_2} is the diffusion coefficient of O₂ in 0.1 M KOH (1.9 $\times 10^{-5}$ cm² s^{−1}), ν is the kinetic viscosity (1.0 $\times 10^{-2}$ cm² s^{−1}) and C_{O_2} is the bulk concentration of O₂ in 0.1 M KOH (1.2 $\times 10^{-6}$ mol cm^{−3}). RRDE measurements were also performed to explore the four-electron sensitivity. The H₂O₂ yield and electron transfer (n) were calculated by following the above equations

$$n = \frac{4j_D}{j_D + \frac{j_R}{N}} \quad (3)$$

$$H_2O_2 \text{ yield (\%)} = \frac{\frac{2j_R}{N}}{j_D + \frac{j_R}{N}} \times 100 \quad (4)$$

where j_D and j_R is the Faradaic current on the disk and ring, respectively. N is hydrogen peroxide collection coefficient of the ring (0.37).

Zn–Air Battery and Overall Water Splitting: The solid-state Zn–air battery was tested with a homemade two-electrode Zn–air cell. First, the GH-BGQD catalyst was mixed with 3 mL of ethanol containing 40 μ L Nafion (5wt%) solution, and the catalyst mixture was then loaded onto Ni foam as the air electrode with a loading of 8 mg cm^{−2}. The polish Zn foil with the thickness of 0.01 mm was used as the anode. The polyvinyl alcohol polymer hydrogel film containing 2.0 M KOH and 0.02 M zinc acetate was used as the solid electrolyte and membrane. The anode, air electrodes, and alkaline polymer gel film were assembled and pressed under pressure of 3 MPa for 5 min using a sheet presser to improve the integrity of laminated structure. For overall water splitting, two-electrode configuration was used in 0.1 M KOH solution, and both anode and cathode electrodes were prepared by the GH-BGQD loaded Ni foams. The loading of catalyst was 8 mg cm^{−2} with the electrode area of 0.5 $\times$ 1.0 cm².

Computational Analysis: Density functional theory calculations were carried out using DMol³ modules in the Materials Studio. Herein, the generalized gradient approximation Perdew–Burke–Ernzerhof (PBE) functional was utilized as well as the double numerical with polarization basis sets. Convergence tolerances of energies and maximum forces of 10^{−5} Ha and 0.002 Ha/Å were respectively chosen for the geometry optimization. The convergence thresholds of maximum displacement and smearing were 0.005 Å and 0.003 Ha, respectively. The interactions of each molecule with the solvent (water) were included using the conductor-like solvent model (COSMO). Dispersion corrections (DFT-D3) were applied and spin polarization was considered in all calculations. To calculate the adsorption energy (E_{ads}) of a hydroxide ion (OH[−]) and a water molecule (H₂O) on various B-GQDs surfaces, the following equation was used

$$E_{\text{ads}} \text{ (eV)} = E_{\text{total}} - E_{\text{OH/H}_2\text{O}} - E_{\text{B-GQDs}} \quad (5)$$

where E_{ads} is the adsorption energy of the adsorbents (i.e., OH[−] and H₂O) on the B-GQDs surface, E_{total} is the total energy of the B-GQDs

surface with the adsorbent, $E_{\text{OH}/\text{H}_2\text{O}}$ is the total energy of a hydroxide ion or water molecule, and $E_{\text{B-GQDs}}$ is the energy of the B-GQD surface.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

graphene hydrogels, graphene quantum dots, trifunctional catalysts, water electrolysis, Zn–air batteries

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